Ho Chi Minh City University of Technology



Faculty of Computer Science and Engineering
Course: Computer Architecture Lab (CO2008)

Lab 3

Advanced instructions

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1 Introduction

- This report presents the solutions for Lab 3 – *Advanced instructions*.
- All programs are written in MIPS assembly and tested on the MARS MIPS simulator.
- Each exercise is verified with three test cases. For every test case, the input, the expected output and a screenshot of the actual run are given.

2 Exercises

2.1 Exercise 1

Requirement

Write a MIPS program to find the solution to the equation:

$$ax^2 + bx + c = 0 \tag{1}$$

where a, b, and c are floating-points number inserted by the user. The result should be in 1 of these 3 cases:

- There are 2 solutions to x: print those results to the screen after the text: "x1 = " and "x2 = ".
- There is 1 solution to x: print the result after the text: "There is one solution, x = ".
- There is no real solution: print "There is no real solution".

Note that the solutions printed must be in floating-point numbers. The user will input the floating-points numbers after the prompts: "Please insert a: ", "Please insert b: ", and "Please insert c: ".

For example, if the user inserted **a=1.2, b=9.6, c=1** then the result should be: **x1 = -7.8944 and x2 = -0.1056**

In another example, if the user inserted **a=2.4, b=-3.2, c=3.1** then the result should be: **There is no real solution**

Test cases

Test case 1

- **Input:** $a = 1.2$, $b = 9.6$, $c = 1$
- **Expected output:** `x1 = -7.89444`, `x2 = -0.10555982`
- **Output:**



Figure 1: Exercise 1 – Test case 1

Test case 2

- **Input:** $a = 1$, $b = -4$, $c = 4$
- **Expected output:** `There is one solution, x = 2.0`
- **Output:**



Figure 2: Exercise 1 – Test case 2

Test case 3

- **Input:** $a = 2.4$, $b = -3.2$, $c = 3.1$
- **Expected output:** `There is no real solution`
- **Output:**

[Screenshot: images/ex1_tc3.png]

Figure 3: Exercise 1 – Test case 3

2.2 Exercise 2

Requirement

Write a MIPS program to calculate the following integral:

$$f(x) = \int_v^u \frac{ax^6 + bx^5 + cx}{d^4 + e^3}\, dx \tag{2}$$

where u, v, a, b, c, d, and e are floating-point numbers chosen by the user. The user will input the floating-point numbers after the prompts: "Please insert a: ", "Please insert b: ", etc. For example, user inserted **u=1, v=2, a=3, b=4, c=5, d=6, e=7** then the result should be: **-0.0634**

Test cases

Test case 1

- **Input:** $u = 1$, $v = 2$, $a = 3$, $b = 4$, $c = 5$, $d = 6$, $e = 7$
- **Expected output:** `Result: -0.063409746`
- **Output:**

[Screenshot: images/ex2_tc1.png]

Figure 4: Exercise 2 – Test case 1

Test case 2

- **Input:** $u = 1$, $v = -1$, $a = 7$, $b = 6$, $c = 2$, $d = 1$, $e = 1$
- **Expected output:** `Result: 1.0`
- **Output:**

[Screenshot: images/ex2_tc2.png]

Figure 5: Exercise 2 – Test case 2

Test case 3

- **Input:** $u = 3$, $v = 0$, $a = 1$, $b = 2$, $c = 3$, $d = 1$, $e = -1$
- **Expected output:** `The denominator d^4 + e^3 is zero, the integral can not be calculated.`
- **Output:**

[Screenshot: images/ex2_tc3.png]

Figure 6: Exercise 2 – Test case 3

2.3 Exercise 3

Requirement

To allocate memory, please refer to the following syscall:

```
    li $v0, 9   # system call code for dynamic allocation
    li $a0, 24  # $a0 contains number of bytes to allocate
```

After the above system call, `$v0` contains the first address in heap memory that is allocated. Then, accessing the allocated memory can be done by `lw`/`sw`, for example:

```
    # Trying to write to allocated space
    addi $t0, $zero, 2021
    sw $t0, 0($v0)
```

The followings are instructions used to access a file (open/close/read/write):

```
    # Sample MIPS program that writes to a new file.
    # by Kenneth Vollmar and Pete Sanderson
    .data
    fout: .asciiz "testout.txt" # filename for output
    msg1: .asciiz "Before read: "
    msg2: .asciiz "After read: "
    buffer_write: .asciiz "The quick brown fox jumps over the lazy
    dog.\n"
    buffer_read: .asciiz "--------------------------------\n"
    .text
    ############################################################
    # Open (for writing) a file that does not exist
    li $v0, 13      # system call for open file
    la $a0, fout    # output file name
    li $a1, 1       # Open for writing (flags are 0: read, 1: write)
    li $a2, 0       # mode is ignored
    syscall         # open a file (file descriptor returned in $v0)
    move $s6, $v0   # save the file descriptor
    ############################################################
    # Write to file just opened
    li $v0, 15      # system call for write to file
    move $a0, $s6   # file descriptor
    la $a1, buffer_write # address of buffer from which to write
    li $a2, 44      # hardcoded buffer length
    syscall         # write to file
    ############################################################
    # Close the file
    li $v0, 16      # system call for close file
    move $a0, $s6   # file descriptor to close
    syscall         # close file
```

```
    ######################################################
    ######################################################
    # Open (for reading) a file
    li $v0, 13       # system call for open file
    la $a0, fout     # input file name
    li $a1, 0        # Open for reading (flags are 0: read, 1: write)
    li $a2, 0        # mode is ignored
    syscall          # open a file (file descriptor returned in $v0)
    move $s6, $v0    # save the file descriptor
    ######################################################
    # Read from file
    li $v0, 14       # system call for read
    move $a0, $s6    # file descriptor
    la $a1, buffer_read # address of buffer read
    li $a2, 44       # hardcoded buffer length
    syscall          # read file
```

Please do the followings:

1. Manually create a line of text in format `<name>,<id>,<address>,<age>,<religion>` in a text file by using a text editor. The file must be named: `raw_input.txt`. An example text should be: `Davy Jones,2251234,168 Ly Thuong Kiet St District 10 Ward 14 HCMC,69,None`
2. Open the file to read the text you inserted.
3. Declare a string in the heap memory with dynamically allocated memory. The size of the string must be large enough to store the text in the text file.
4. Copy the line from the text file to the string in the memory.
5. Print the string to the terminal in the following format:
   (a) -----Student personal information-----
   (b) Name: <name>
   (c) ID: <ID>
   (d) Address: <address>
   (e) Age: <age>
   (f) Religion: <religion>

   The output in the terminal for the former example input should be:

```
-----Student personal information-----
Name: Davy Jones
ID: 2251234
Address: 168 Ly Thuong Kiet St District 10 Ward 14 HCMC
```

```
Age: 69
Religion: None
```

6. The printed result should also be written in a file named `formatted_result.txt`

Test cases

For each test case, `raw_input.txt` is placed in the working directory of MARS. After the run, the content of `formatted_result.txt` is the same as the terminal output.

Test case 1

- **Input:** `raw_input.txt` contains
  ```
  Davy Jones,2251234,168 Ly Thuong Kiet St District 10 Ward 14 HCMC,69,None
  ```
- **Expected output:**
  ```
  -----Student personal information-----
  Name: Davy Jones
  ID: 2251234
  Address: 168 Ly Thuong Kiet St District 10 Ward 14 HCMC
  Age: 69
  Religion: None
  ```
- **Output:**

[Screenshot: images/ex3_tc1.png]

Figure 7: Exercise 3 – Test case 1

Test case 2

- **Input:** `raw_input.txt` contains
  `Nguyen Van An,2512345,268 Ly Thuong Kiet St Ward 14 District 10 HCMC,19,Buddhism`
- **Expected output:**
  `-----Student personal information-----`
  `Name: Nguyen Van An`
  `ID: 2512345`
  `Address: 268 Ly Thuong Kiet St Ward 14 District 10 HCMC`
  `Age: 19`
  `Religion: Buddhism`
- **Output:**

[Screenshot: images/ex3_tc2.png]

Figure 8: Exercise 3 – Test case 2

Test case 3

- **Input:** `raw_input.txt` contains (the line ends with a Windows line break `\r\n`)
  `Tran Thi Bich Ngoc,2310001,1 Vo Van Ngan St Linh Chieu Ward Thu Duc City,21,Christianity`
- **Expected output:**
  `-----Student personal information-----`
  `Name: Tran Thi Bich Ngoc`
  `ID: 2310001`
  `Address: 1 Vo Van Ngan St Linh Chieu Ward Thu Duc City`
  `Age: 21`
  `Religion: Christianity`
- **Output:**

[Screenshot: images/ex3_tc3.png]

Figure 9: Exercise 3 – Test case 3